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Ridge regression

Ridge regression in a form of linear regression which prefers coefficients of smaller magnitude when fitting training data.

Why might shrinking regression coefficients be desirable in the first place??

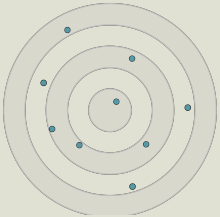
Example. Two researchers, working independently, each model a response variable y using two explainers, x_1 and x_2. Unbeknownst to the researchers, each of the predictors has equal explanatory power.

Not knowing the true relationship between the variables, the researchers each collect a random sample and fit a simple linear regression model. They find:

$$\text{Researcher 1:} \quad \hat{y}_i = 2.47 + 1.26x_{1i} + 0.74x_{2i}$$
$$\text{Researcher 2:} \quad \hat{y}_i = 2.49 + 0.77x_{1i} + 1.27x_{2i}$$

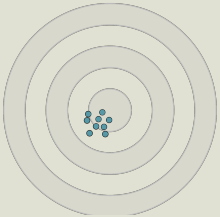
As a result, they draw different conclusions about the roles of x_1 and x_2 in the process they're investigating.

The difference between the two models is due to **model variability**, which arises from the random process used to obtain the samples.



Although the modeling procedure may give good results on average, any single model is highly unreliable.

Ridge regression attempts to reduce model variability by introducing a small amount of bias.



An individual model may perform very well even though the process that created it would do worse than OLS regression, on average.

Recall that a linear regression model $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$ is fit by minimizing the sum of the squared residuals across all possible values of the coefficients $\beta_0, \ldots, \beta_p$.

$$\mathrm{RSS} = \sum_i (\hat{y}_i - y_i)^2$$

Here y_i is the observed value of the response variable while $\hat{y}_i$ is the value predicted by the model.

Ridge regression minimizes a slightly different quantity.

$$\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

The first part of this expression is the same residual sum of squares used in simple linear regression. The second part represents a penalty on larger coefficient values, which has the effect of shrinking their optimal values somewhat.

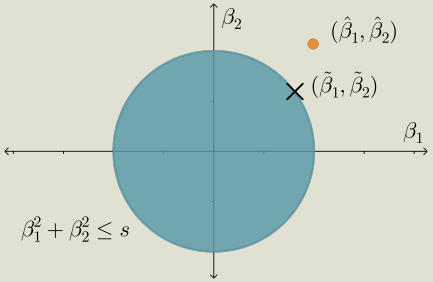
Ridge regression is closely related to **the lasso**, which minimizes a very similar quantity.

$$\text{Ridge:} \quad \sum_{i=1}^{n}(\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

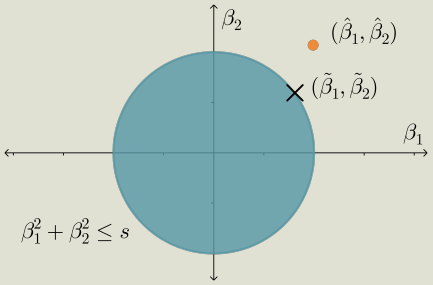
$$\text{Lasso:} \quad \sum_{i=1}^{n}(\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

Because λ isn't determined directly by the training data, it's referred to as a **hyperparameter**. An optimal value is typically chosen using cross-validation.

Ridge regression can be viewed as minimizing RSS among coefficient values within a specified distance of the origin. The distance s is uniquely determined by λ.



A penalty of $\lambda = 0$ (ordinary least squares regression) corresponds to a region of infinite radius. As λ increases, that radius gets smaller.



Typically, test error decreases even after the allowable set no longer includes $(\hat{\beta}_1, \hat{\beta}_2)$ but soon begins to increase again.

Once a value of λ has been chosen, maximum likelihood estimation yields coefficients $\tilde{\beta}_0, \ldots, \tilde{\beta}_p$ just as it does with OLS regression. The result is again a linear model,

$$y_i = \tilde{\beta}_0 + \tilde{\beta}_1 x_{i1} + \cdots + \tilde{\beta}_p x_{ip}.$$

We can interpret the coefficients as rates of change, just as we would if no ridge penalty had been imposed.

Let's return to the example from earlier, where different random samples led to OLS regression models with very different coefficients β_1 and β_2.

$$\text{Researcher 1:} \quad \hat{y}_i = 2.47 + 1.26x_{1i} + 0.74x_{2i}$$
$$\text{Researcher 2:} \quad \hat{y}_i = 2.49 + 0.77x_{1i} + 1.27x_{2i}$$

In this scenario, a ridge regression might lead to the following results:

$$\text{Researcher 1:} \quad \hat{y}_i = 2.49 + 0.97x_{1i} + 0.95x_{2i}$$
$$\text{Researcher 2:} \quad \hat{y}_i = 2.51 + 0.92x_{1i} + 0.91x_{2i}$$

The models are now in alignment.

The ridge penalty, $\lambda \sum \beta_j^2$, is harsher toward variables with larger values, which means it's sensitive to the relative scales of the variables.

For this reason, it's standard practice to standardize each of the variables prior to carrying out a ridge regression. This places them on a common scale and ensures that the ridge penalty considers them all equally.

Ridge regression is helpful when a model has high variance, typically because at least one of the following is true:

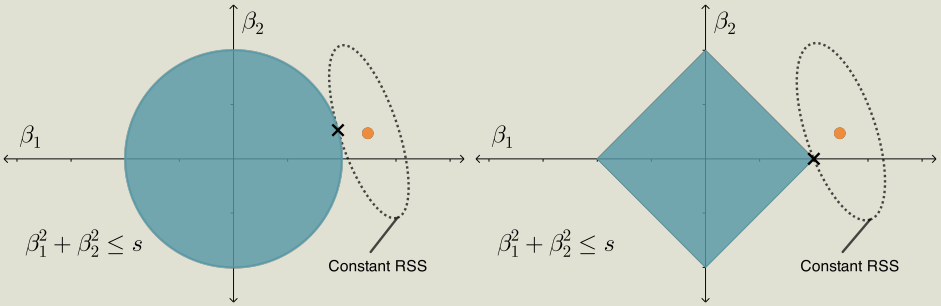
- The number of predictor variables is large relative to the number of observations.
- The predictors are highly correlated with one another.

Each of these can be checked directly in the training set.

Unlike the lasso, ridge regression never shrinks any coefficient all the way to zero.

When two predictors are correlated with one another, the ridge penalty tends to push their coefficients toward one another, taking both of them into account. The lasso penalty, on the other hand, tends to pick one explainer and zero out the other.

This behavior has a geometric explanation.



Use this differences to decide whether to use lasso or ridge regression.

- Use the lasso when variable selection is desired, typically because the most important variables are highly correlated with less relevant ones.
- Use ridge regression when you expect all variables to be relevant, even if their individual impacts are small.

Alternatively, you can opt for an **elastic net**, which uses both penalties simultaneously, weighting the two with an additional tuning parameter.

